

OpenCPU SDK

HTTP API Notes

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Notice

This document is specifically for **N706B** and **N520** modules.

This document is intended for system engineers (SEs), development engineers, and test engineers.

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About This Document

Scope

This document is applicable to N706B and N520 modules.

Audience

This document is intended for [system engineers \(SEs\)](#), [development engineers](#), and [test engineers](#).

Change History

Issue	Date	Change	Changed By
1.0	2023-05	Initial release	Gao Zheng

Conventions

Symbol	Indication
	This warning symbol means danger. You are in a situation that could cause fatal device damage or even bodily damage.
	Means reader be careful. In this situation, you might perform an action that could result in module or product damages.
	Means note or tips for readers to use the module

1 Overview

This document introduces HTTP-related APIs in OpenCPU SDK, focusing on the following aspects:

- param Information
Used for setting HTTP server, port, callback functions, etc.
- HTTP APIs
Used to establish connections, send requests, and close connections.

2 Data Structure

2.1 nwy_http_handle_t

```
typedef void *nwy_http_handle_t;
```

2.2 nwy_http_event_e

```
typedef enum
{
    NWY_HTTP_DNS_ERR,          //Dns error
    NWY_HTTP_OPEN_FAIL,       //open fail
    NWY_HTTP_OPENED,          //setup success
    NWY_HTTPS_SSL_CONNECTED,   //https setup success
    NWY_HTTP_CLOSED_PASV,     //server closed
    NWY_HTTP_CLOSED,          //client closed
    NWY_HTTP_DATA_RECVED,     //recv data
    NWY_HTTP_DATA_SEND_ERR,   //data send error
    NWY_HTTP_EVENT_SSL_CONNECT_FAIL,
    NWY_HTTP_DATA_SEND_FINISHED, //data send finished
    NWY_HTTP_EVENT_UNKOWN,    //unkown error
}nwy_http_event_e;
```

2.3 nwy_http_content_type_e

```
typedef enum
{
    NWY_HTTP_CONTENT_TYPE_URLENCODE, //"Content-Type: application/x-www-form-urlencoded",
    NWY_HTTP_CONTENT_TYPE_TEXT, //"Content-Type: application/text",
    NWY_HTTP_CONTENT_TYPE_JSON, //"Content-Type: application/json",
    NWY_HTTP_CONTENT_TYPE_XML, //"Content-Type: application/xml",
    NWY_HTTP_CONTENT_TYPE_HTML, //"Content-Type: application/html",
}nwy_http_content_type_e;
```

2.4 nwy_http_param_t

```
typedef struct{
    unsigned short cid;
    int timeout_s;
    char *host;
```

```
int port;
nwy_http_resultcb cb;//callback function
}nwy_http_param_t;
```

2.5 http_request_param_t

```
typedef struct
{
    unsigned int keepalive;
    nwy_http_content_type_e type;
    const char* data;
    int len;
    char * uri; //Resource identifier, resource path begins with backslash '/', for example:
    '/boot/test.txt'
}http_request_param_t;
```

2.6 http_get_param_t

```
typedef struct
{
    unsigned char keepalive;
    int offset;
    int len;
    char * uri; Resource identifier, the resource path begins with a backslash '/', for
    example: '/boot/test.txt'
}http_get_param_t;
```

2.7 http_head_param_t

```
typedef struct
{
    unsigned char keepalive;
    char * uri; Resource identifier, the resource path begins with a backslash '/', for
    example: '/boot/test.txt'
}http_head_param_t;
```

2.8 http_put_param_t

```
typedef http_request_param_t http_put_param_t;
```

2.9 http_post_param_t

```
typedef http_request_param_t http_post_param_t;
```

2.10 Callback Function Definition

```
typedef struct{
    unsigned short cid; //data channel
    nwy_http_handle_t http_handle; //HTTP client handle.
    unsigned int event; /* Event id */
    unsigned int data_len; /*Get the length of the received data */
    char* data; /* Acquired data content */
}nwy_http_result_t;
typedef void (*nwy_http_resultcb)(nwy_http_result_t *event);
```


3 APIs

The HTTP API definitions can be found in `nwy_http.h`.

3.1 `nwy_http_setup`

Format	<code>nwy_http_handle_t nwy_http_setup(nwy_http_param_t http_param, nwy_ssl_conf_t *ssl_cfg)</code>
Description	Establishes an HTTP/SSL HTTP link.
Parameters	<code>nwy_http_handle_t handle</code> : HTTP/SSL HTTP client handle. <code>http_param</code> : <code>cid</code> : Channel range 1–7. One channel supports up to 3 HTTP clients. <code>char *url</code> : Path to the resource. <code>int port</code> : Target port number. <code>nwy_http_resultcb cb</code> : Event notification callback function. <code>int timeout</code> : HTTP connection timeout in seconds. <code>nwy_ssl_conf_t *ssl_cfg</code> : SSL configuration details. Refer to <code>nwy_ssl_config.h</code> . NULL indicates HTTP, while non-NULL indicates SSL HTTP encryption.
Return Value	Success: Returns HTTP/SSL HTTP client handle. Failed: NULL

3.2 `nwy_http_get`

Format	<code>nwy_error_e nwy_http_get(nwy_http_handle_t handle, http_get_param_t *param);</code>
Description	Initiates a GET request.
Parameters	<code>nwy_http_handle_t handle</code> : HTTP/SSL HTTP client handle. <code>http_get_param_t *param</code> : GET parameters: unsigned char keepalive (0, 1): 0: Short connection 1: Long connection int offset: Offset to specify the download starting position. int len: Length to specify the download size. char *uri: Resource identifier, with the path starting with /, e.g., "/boot/test.txt".
Return Value	Success: <code>NWY_SUCCESS</code>

Remarks	Failed: Check nwy_error_e
	• If offset and len are 0, the entire package is downloaded. • The total package size can be obtained via the nwy_http_get_head() interface. • A non-zero len indicates a breakpoint request for downloading.

3.3 nwy_http_head

Format	nwy_error_e nwy_http_head(nwy_http_handle_t handle, http_head_param_t *param)
Description	Initiates a HEAD request to retrieve resource header information.
Parameters	nwy_http_handle_t handle: HTTP/SSL HTTP client handle. http_head_param_t *param: HEAD parameters: unsigned char keepalive (0, 1): 0: Short connection 1: Long connection char *uri: Resource identifier, with the path starting with /, e.g., "/boot/test.txt".
Return Value	Success: NWY_SUCCESS Failed: Check nwy_error_e

3.4 nwy_http_post

Format	nwy_error_e nwy_http_post(nwy_http_handle_t handle, http_post_param_t *param)
Description	Initiates a POST request to modify a resource.
Parameters	nwy_http_handle_t handle: HTTP/SSL HTTP client handle. http_post_param_t *param: http post parameters: unsigned char keepalive (0, 1): 0: Short connection 1: Long connection nwy_http_content_type_e: Content type of the message. const char *data: POST data, with a maximum length of 3072 bytes. int len: Data length. char *uri: Resource identifier, with the path starting with /, e.g., "/boot/test.txt".
Return Value	Success: NWY_SUCCESS Failed: Check nwy_error_e
Remarks	Maximum upload data size is 3072 bytes. For larger uploads, use the nwy_http_open_mode() interface.

3.5 nwy_http_put

Format	nwy_error_e nwy_http_put(nwy_http_handle_t handle, http_put_param_t *param)
Description	Initiates a PUT request to create a resource.
Parameters	nwy_http_handle_t handle: HTTP/SSL HTTP client handle. http_put_param_t *param: PUT parameters: - unsigned char keepalive (0, 1): 0: Short connection 1: Long connection - const char *data: Resource data, with a maximum length of 3072 bytes. - int len: Data length. nwy_http_content_type_e type: Content type of the message. char *uri: Resource identifier, with the path starting with /, e.g., "/boot/test.txt".
Return Value	Success: NWY_SUCCESS Failed: Check nwy_error_e
Remarks	Maximum upload data size is 3072 bytes. For larger uploads, use the nwy_http_open_mode() interface.

3.6 nwy_http_open_mode

Format	nwy_error_e nwy_http_open_mode(nwy_http_handle_t handle, unsigned char keepalive, const char* data, int len)
Description	Initiates a custom message request.
Parameters	nwy_http_handle_t handle: HTTP/SSL HTTP client handle. unsigned char keepalive (0, 1): 0: Short connection 1: Long connection const char * data, custom message data unsigned char len, data length
Return Value	Success: NWY_SUCCESS Failed: Check nwy_error_e
Remarks	Add custom HTTP headers and body when using this mode. Suitable for handling large data and custom HTTP headers.

3.7 nwy_http_close

Format	nwy_error_e nwy_http_close(nwy_http_handle_t handle)
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Description	Actively closes an HTTP/HTTPS link.
Parameters	nwy_http_handle_t handle: HTTP/SSL HTTP client handle.
Return Value	Success: NWY_SUCCESS Failed: Check nwy_error_e